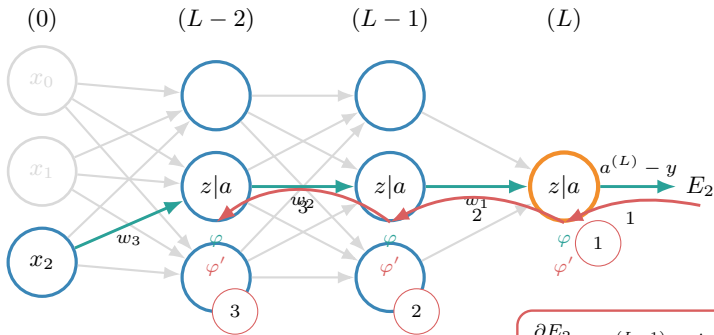


Gradient descent | partial derivative



$$\frac{\partial E_2}{\partial w_1} = a^{(L-1)} \cdot \varphi'(z^{(L)}) \cdot 1$$

$$\frac{\partial E_2}{\partial w_2} = a^{(L-2)} \cdot \varphi'(z^{(L-1)}) \cdot w_1 \cdot \varphi'(z^{(L)}) \cdot 1$$

$$\frac{\partial E_2}{\partial w_3} = a^{(L-3)} \cdot \varphi'(z^{(L-2)}) \cdot w_2 \cdot \varphi'(z^{(L-1)}) \cdot w_1 \cdot \varphi'(z^{(L)}) \cdot 1$$